

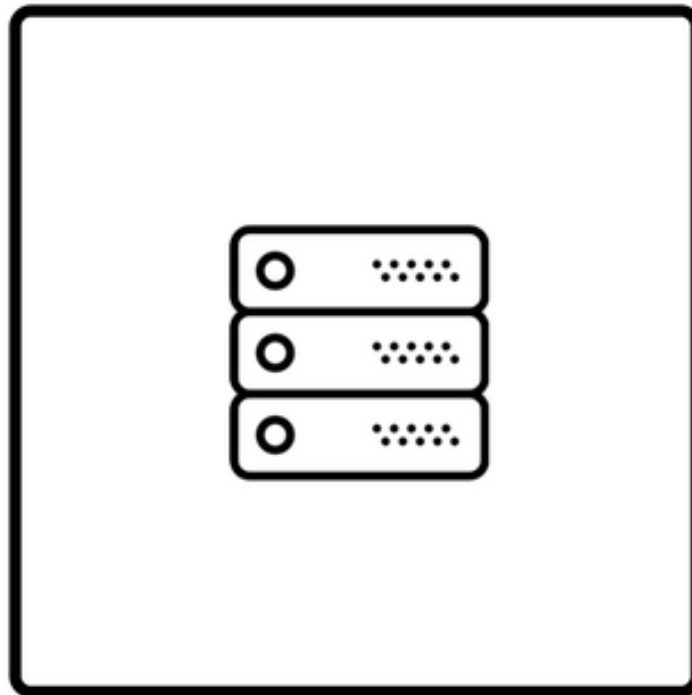
Computação Serverless

Por Rodrigo da Rosa Righi

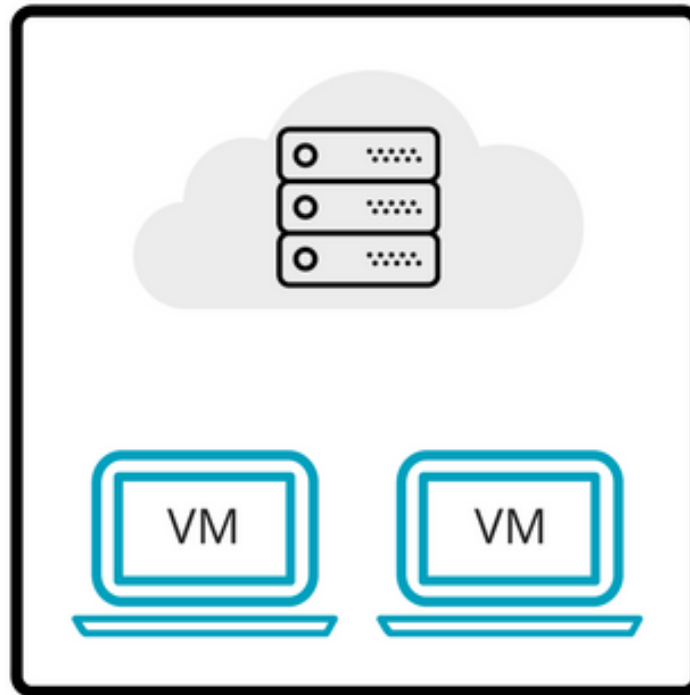
rrrighi@unisinos.br

Evolution of Cloud

Pre-Cloud



Cloud 1.0



Serverless



SERVERLESS

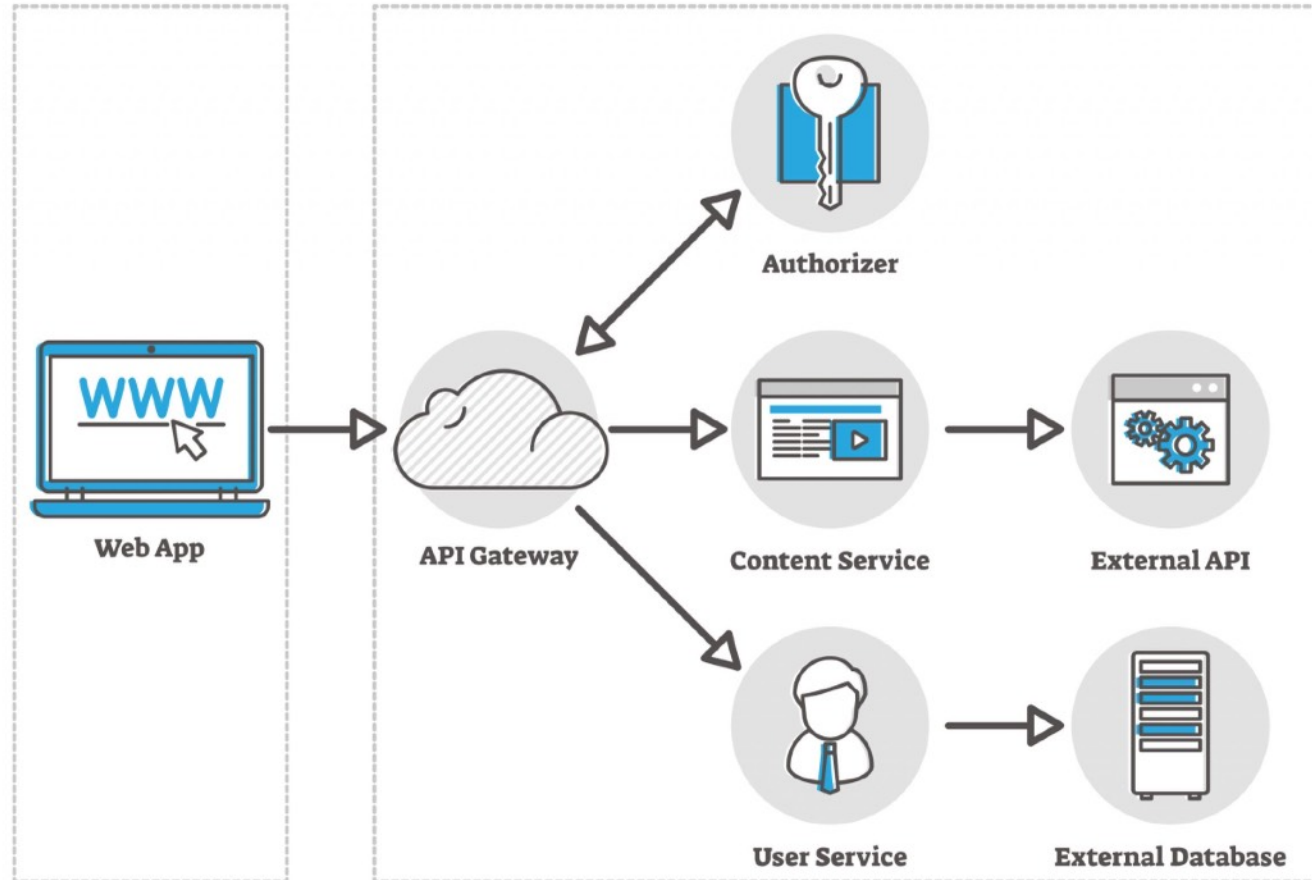
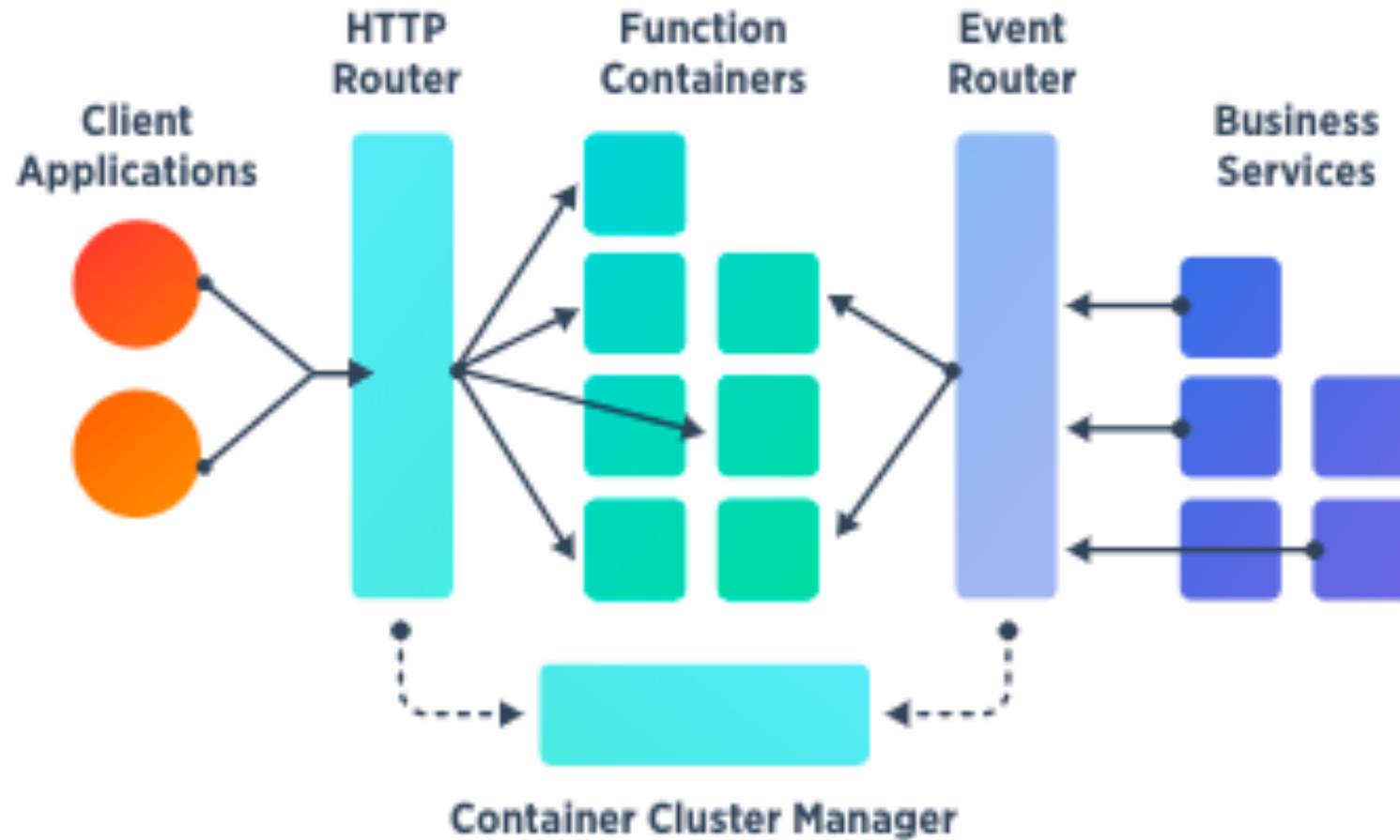
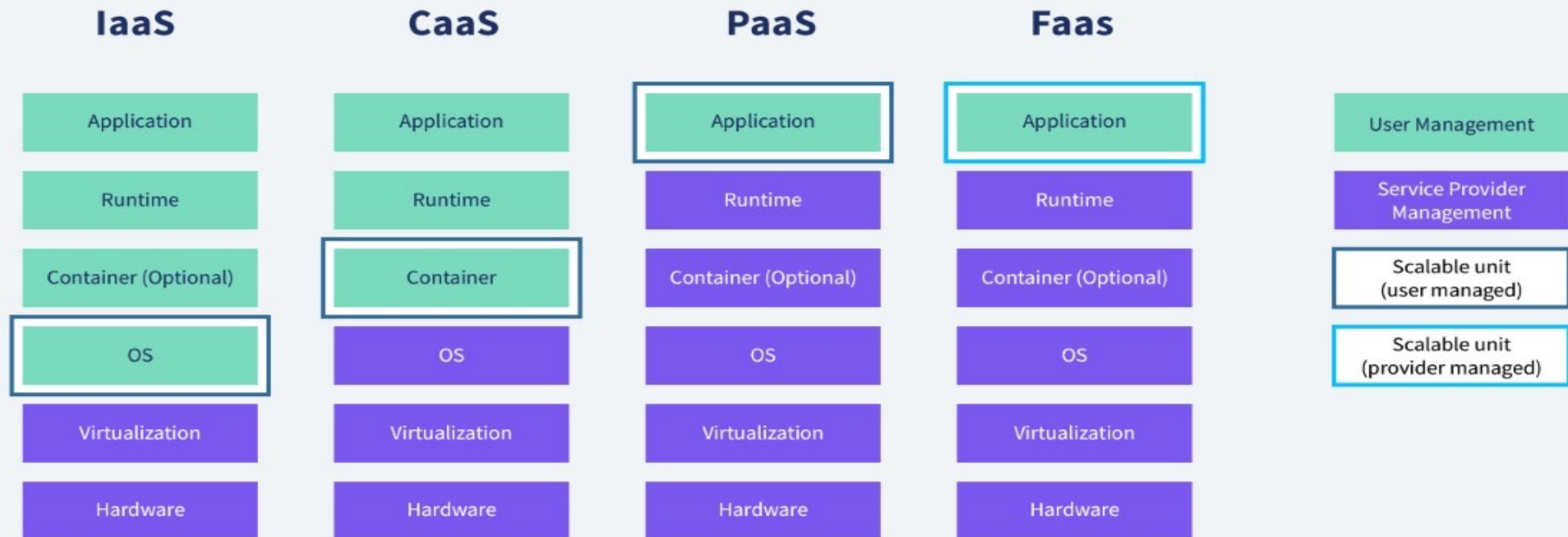


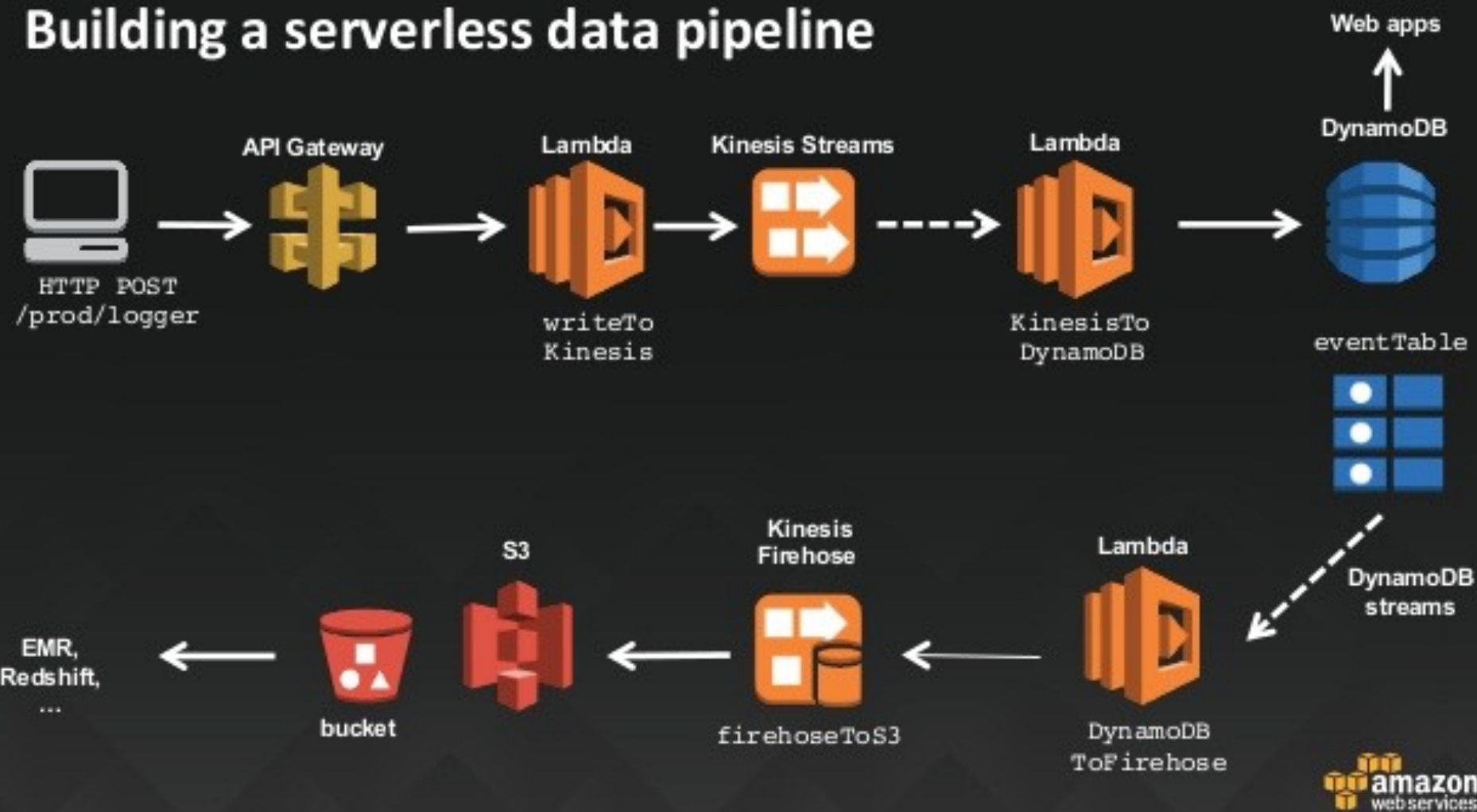
FIGURE 3: SERVERLESS ARCHITECTURE OVERVIEW



SERVERLESS BUSINESS BENEFITS



Building a serverless data pipeline





Benefits of Serverless

Effortless efficiency

Productivity boost

Cost-effective

Simplified operations

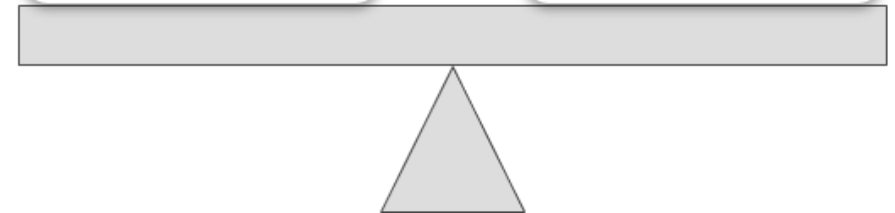
Drawbacks of Serverless

Debugging

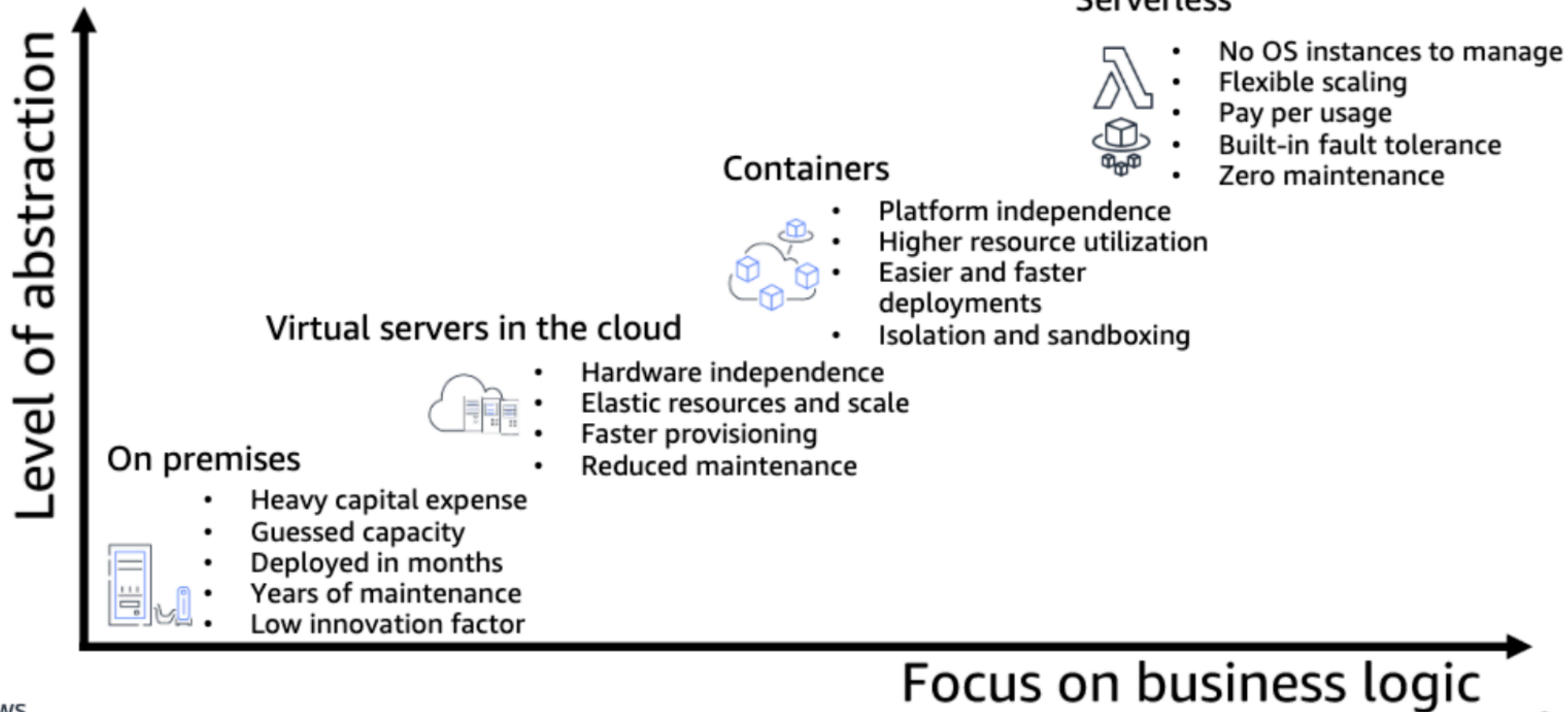
Security

Vendor limitations

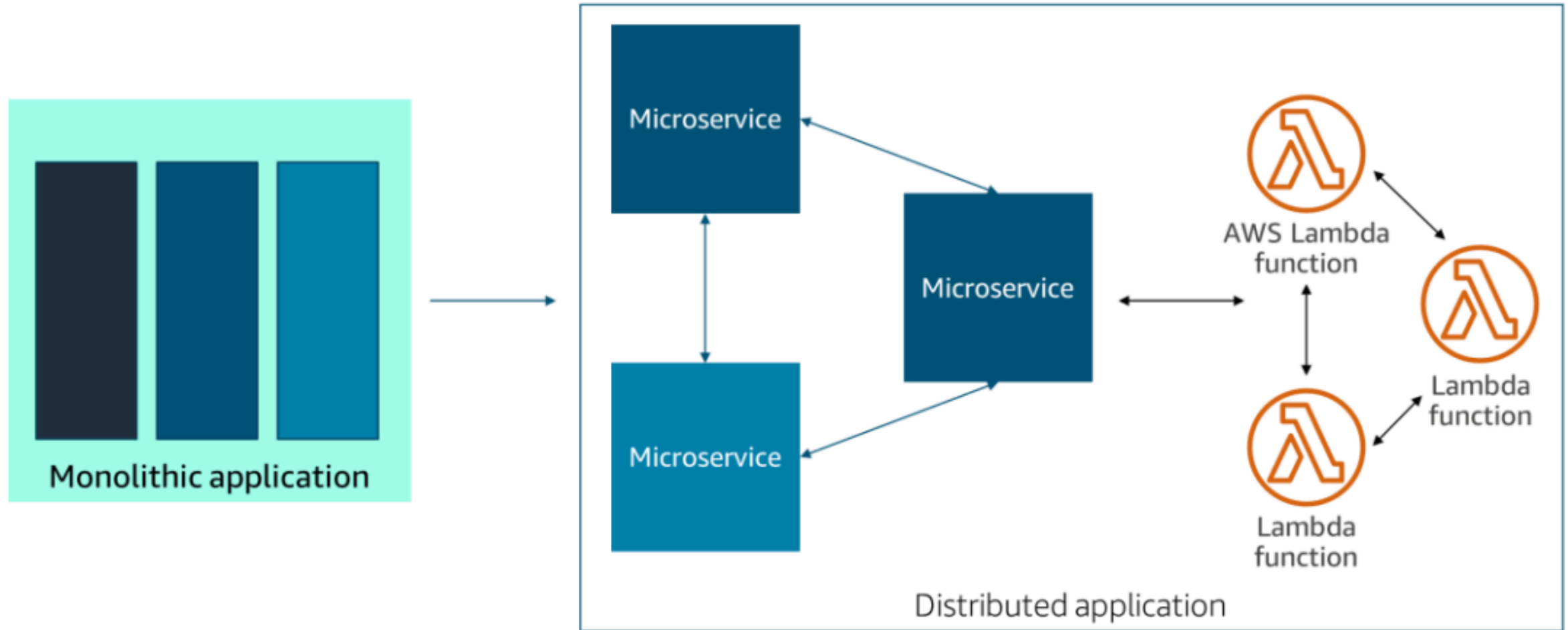
Response latency



Comparison criteria	Serverless	Containerization	Virtualization
Quick start	Instant	Fast	Slow
Resource consumption	Low	Medium	High
Vertical scalability	Automatic	Scalable	Manual/expense
Horizontal scalability	Scalable	Limited	Limited
Operation	Local	Shared	Separate
Operational issues	Low	Moderate	Complex
Configuration capability	Limited	Medium	Abundant
Transferability	Easy	Easy	Medium
Flexibility	Very High	High	Good
Platform limitations	None	Limited	Limited
Cost	Inexpensive	Inexpensive	Expensive
Management	Easy	Medium	Complex
Support time	Medium	Medium	Long
Cost model	Fixed resources	Fixed resources	Pay-per-use model
Security	Secure	Separate	Separate
Reconfiguration required	Low	Low	Moderate
Isolation	Low level	High level	Full



Microservices architecture for distributed applications





Back4App



Parse



Kinvey



AWS Lambda



Azure Functions

Serverless Hosting Providers



Cloudflare Workers



Google Cloud
Functions



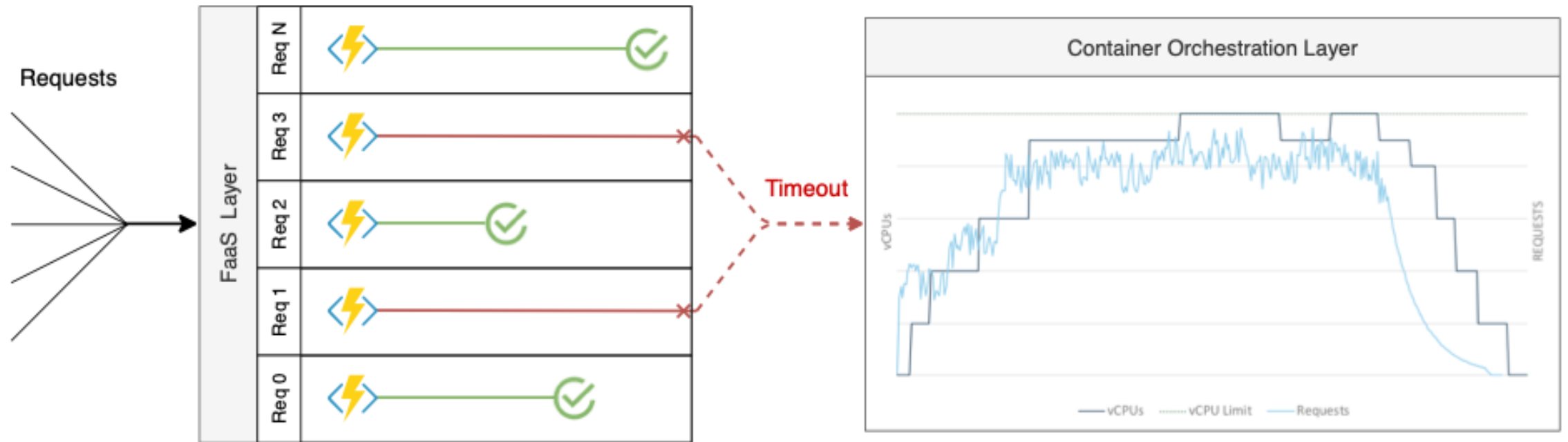
IBM Cloud
Functions

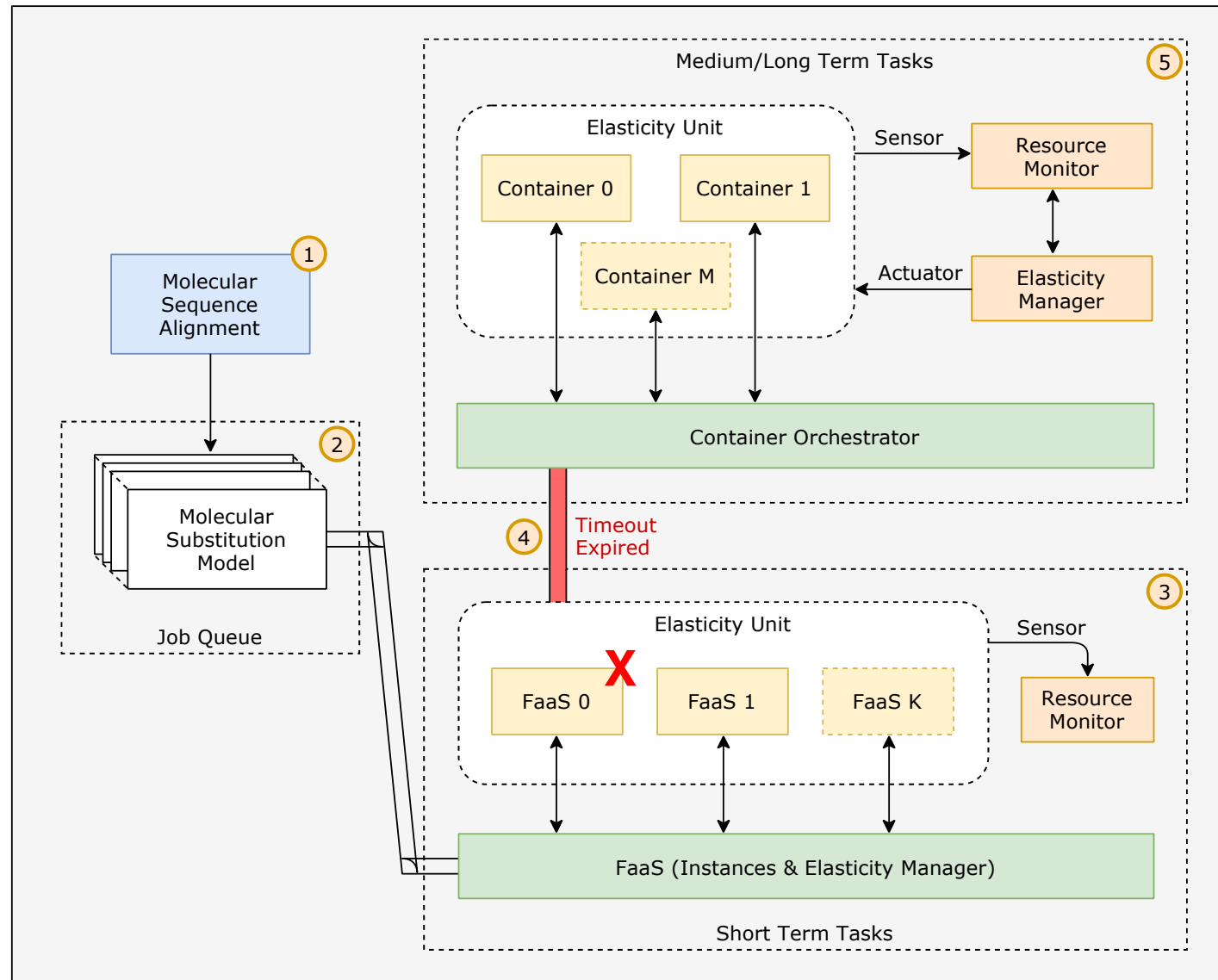


Oracle Functions

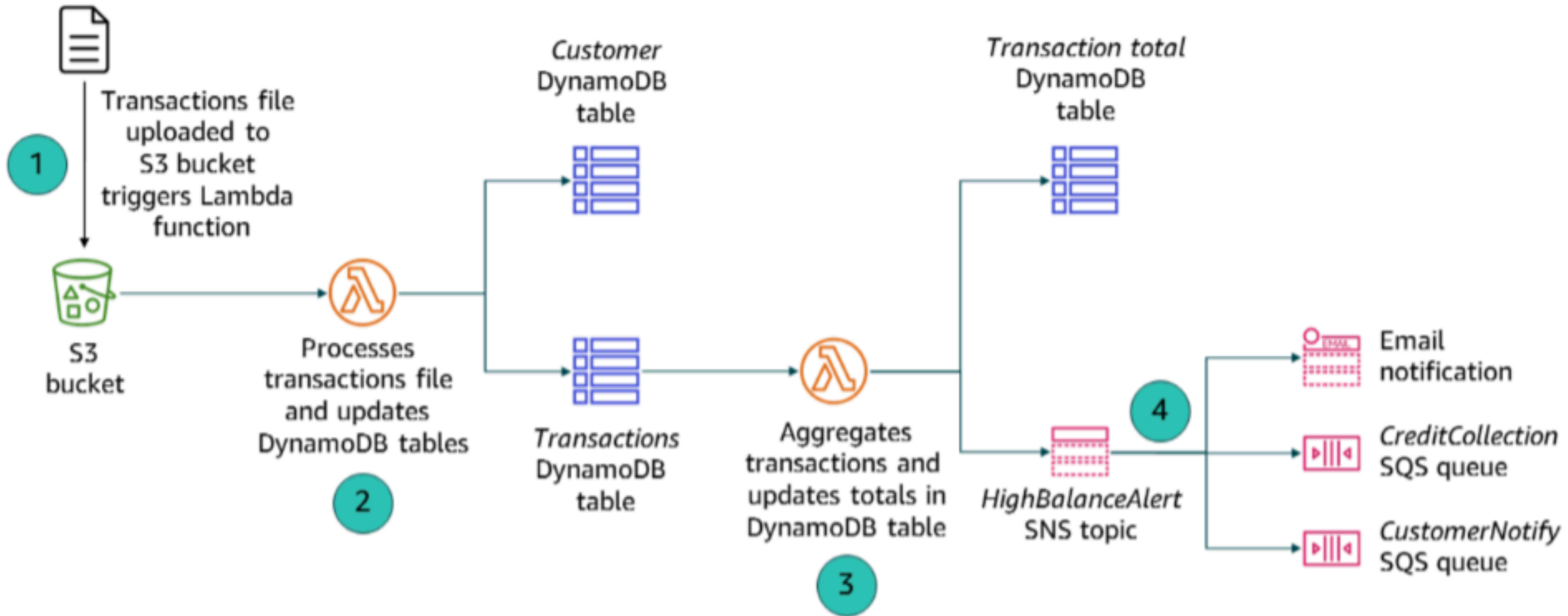


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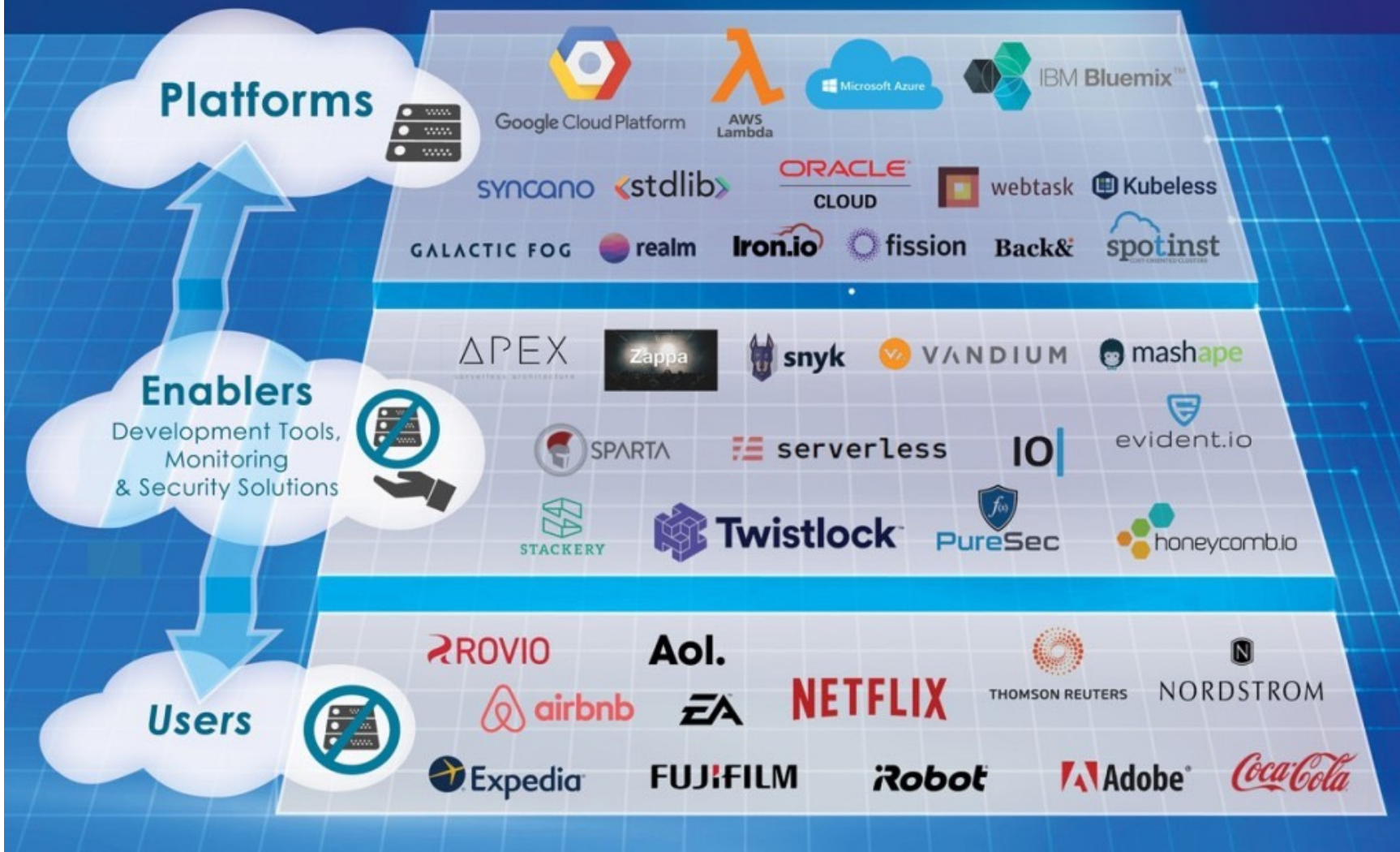




Event-based Lambda function example: Order processing



The State of Serverless



Computação Serverless

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